

Automated Analyst Report

Generated: July 05, 2026 20:07

Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

****EXECUTIVE SUMMARY**** Haircare products exhibit the highest average defect rate (2.48 %) while cosmetics are the cleanest line (1.92 %). Carrier C delivers the strongest revenue-to-shipping-cost performance (US\$ 1,138.56 per shipping dollar). A total of 36 SKUs failed inspection, with the most severe defects on SKU42 (4.94 %). Stock-out exposure is acute for a handful of items, notably SKU34 (risk score 13) which has only one unit on hand against a 26-day lead time.

****KEY FINDINGS**** - ****Product-type defect hierarchy:**** Haircare = 2.483 % > Skincare = 2.335 % > Cosmetics = 1.919 % (Avg Defect Rate by Product Type). - ****Carrier efficiency:**** Carrier C generates the highest revenue per shipping dollar (US\$ 1,138.56), outperforming Carrier B (US\$ 1,055.70) and Carrier A (US\$ 917). - ****Failed-inspection SKUs:**** 36 SKUs flagged; top three defect rates are SKU42 (4.939 %), SKU65 (4.911 %), and SKU50 (4.755 %). All three are linked to Carrier C (Road/Rail) and have limited stock (22, 57, 31 units respectively). - ****Stock-out risk – highest scores:**** - SKU34 (Skincare) – stock = 1, lead time = 26 days, risk score = 13. - SKU68 (Haircare) – stock = 0, lead time = 8 days, risk score = 8. - SKU24 (Haircare) – stock = 4, lead time = 26 days, risk score = 5.2. - ****Margin resilience:**** Gross-margin ratios remain above 99 % for all categories (Cosmetics 0.993, Haircare 0.991, Skincare 0.992), indicating pricing strength despite quality variances. - ****Transportation-mode impact:**** Air shipments have the lowest average defect rate (1.824 %) across 26 SKUs, while Road shows the highest (2.621 % over 29 SKUs).

****RISKS AND RECOMMENDATIONS**** The elevated defect rate in haircare and the concentration of failed-inspection SKUs on Carrier C pose quality-reliability risks; prioritize corrective action with Carrier C and consider shifting high-risk haircare lines to carriers with lower defect profiles (e.g., Carrier B). Immediate inventory replenishment is required for the top stock-out risk SKUs—especially SKU34 and SKU68—to mitigate service disruptions, and a review of lead-time contracts should be undertaken to align supply with demand.